

CTF Writeup: Telemetry

Challenge Name: Telemetry

Category: Forensics / Geospatial

Flag: `SK-CERT{MY_QU4D_W45_H1J4CK3D}`

1. Challenge Overview

This forensics challenge tasks analysts with investigating a MAVLink drone log to uncover a hidden flag. While the file contains decoy strings and intentionally broken metadata, the true flag is encoded geographically within the drone's flight coordinates.

- **File:** telemetry.data (MAVLink 2.0 Binary)
- **Context:** Suspicious drone flight over a "Wheat Field."
- **Points of Interest:** 477 unique GPS points extracted from the log.

The "Honeytoken" Trap: A simple string analysis reveals a decoy flag: `flag{telemetry_payloads_are_a_trap}`. This is placed in the STATUSTEXT or GCS message fields to mislead analysts.

2. Protocol Analysis

The file uses the MAVLink 2.0 protocol. By parsing the binary structure (starting with the magic byte `0xfd`), we can isolate **Message ID #33: GLOBAL_POSITION_INT**.

Each Message #33 packet contains the core data needed to reconstruct the drone's path:

- **Latitude (lat):** Scaled by 10^7 .
- **Longitude (lon):** Scaled by 10^7 .
- **Altitude (alt):** Millimeters above MSL.

3. Data Extraction & Filtering

Using a custom Python script, all unique latitude and longitude pairs were extracted. Analysis of these coordinates reveals two distinct clusters:

- **The Decoy Cluster (High Longitude):** Coordinates at a higher altitude/longitude that trace out a distraction pattern.
- **The Flag Cluster (Lower Longitude):** A dense set of 474 unique points located in the "wheat field" area.

By calculating the mid-longitude of the entire set, the decoy data was filtered out to isolate the flag.

4. Visual Reconstruction

Standard terminal outputs often suffer from "aspect ratio distortion" (characters are taller than they are wide). By mapping the coordinates to a high-resolution 2D scatter plot using a tool like `matplotlib` or a normalized grid, the flight path transforms into legible characters.

The drone was essentially used as a "digital pen" to trace characters in the field.

5. Flag Discovery

The 474 unique points in the wheat cluster trace out the following message:

6. Tools & Techniques Summary

- **Binary Parsing:** Custom Python logic using `struct.unpack` to read raw int32 GPS fields.
- **MAVLink 2.0 Awareness:** Correctly handling the 10-byte header and optional 13-byte signature to maintain packet synchronization.
- **Geospatial Visualization:** Plotting X (Longitude) and Y (Latitude) to reveal hidden text.
- **Coordinate Normalization:** Using column-grouping logic to verify character segments in distorted terminal views.